Multi-Agent Architecture & Pipeline Flow

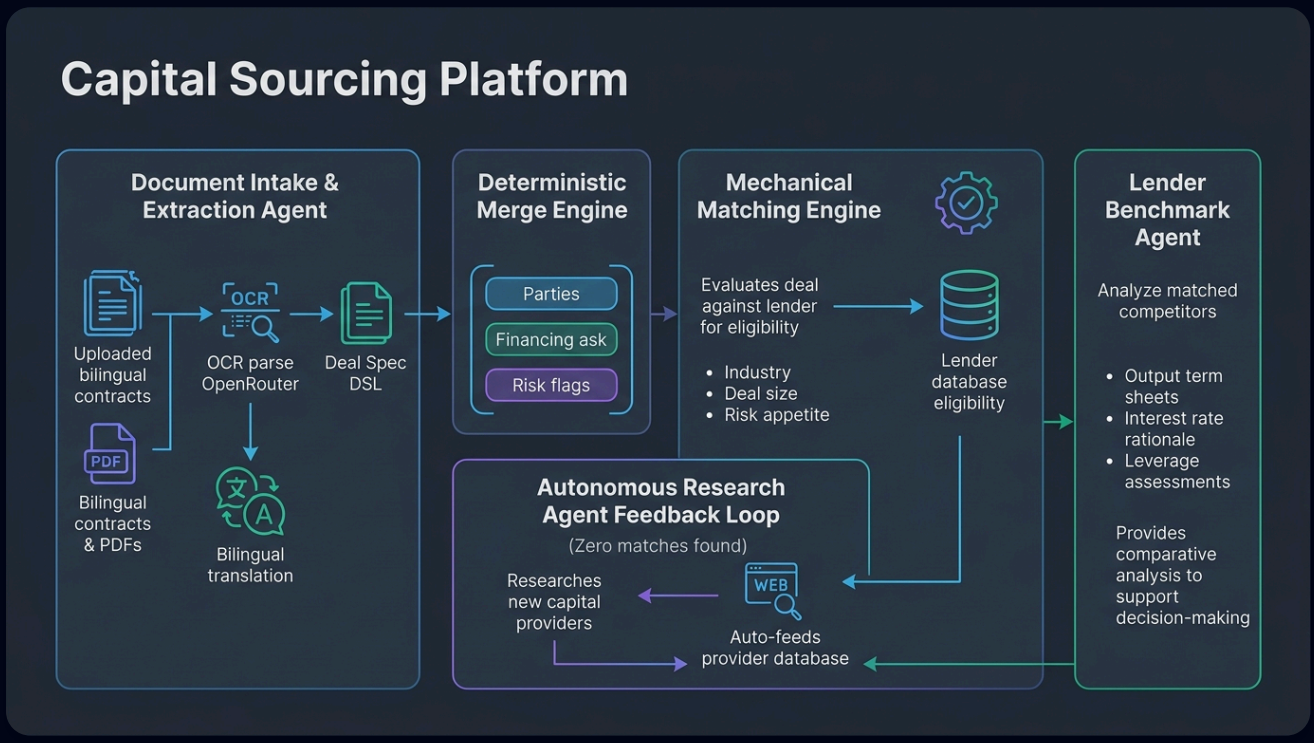
Capital Sourcing Platform

How autonomous extraction, deterministic merging, mechanical matching, dynamic research, and lender benchmarking coordinate.

⚡ Micro-Requests (<60s Safe)

System Interaction Diagram

Generated Architecture Visual



STEP 1 • INTAKE

`DeepSeek-v4-Flash`

Extraction Agent

Processes uploaded bilingual contracts, mining IUPs, invoices, and teasers one by one.

- **OCR Engine:** Mistral-OCR via OpenRouter universal file parser.
- **Taxonomy:** Classifies against Standard Document Taxonomy.
- **Output:** Strict Zod `DocumentExtraction` schema.

STEP 2 • RECONCILIATION

`Deterministic Code`

Spec Merge Engine

Fast in-memory database consolidation. No black-box LLM hallucination for core financial math.

- **Entity Deduplication:** Consolidates parties (Borrower, Obligor, Intermediary).
- **Conflict Detection:** Flags quantity & jurisdiction discrepancies as `RiskFlag`.
- **Readiness Check:** Gates matching if chain-of-title or structures are unknown.

STEP 3 • SCREENING

`Mechanical Filter`

Matching Engine

Screens deals against the global Capital Provider database using verifiable criteria.

- **4 Hard Checks:** Seller jurisdiction, obligor jurisdiction, ticket bounds, & structure.
- **Freshness:** Alerts if provider data > 90 days stale.
- **Outcome:** Generates `ProviderMatch` rows with status & checklist.

STEP 4 • AUTO FEEDBACK LOOP

`Qwen3-235B + Exa Web Search`

Autonomous Research Agent

Triggers if 0 Eligible Matches

If the database yields zero paper-eligible matches, the platform autonomously launches web intelligence rather than dead-ending:

Part A: Web Discovery

Executes Exa-augmented queries tailored to the deal's structure, ticket size, and country corridor.

Part B: Strict Schema Upsert

Validates findings against `ProviderSchema` with strict citation requirements; auto-rematches immediately.

STEP 5 • INTELLIGENCE

`Qwen3-30B-Instruct`

Lender Benchmark Agent

On-demand synthesis for lenders to price their risk and structure competitive term sheets.

- **Pricing Power:** Assesses lender leverage against alternative options.
- **Term Sheet:** Recommends interest rate spread, advance rate %, & tenors.
- **Conditions Precedent:** Identifies security packages & risk mitigants.

The Closed-Loop Feedback Flow

